PRASAD V. POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY

(Autonomous)
KANURU, VIJAYAWADA-520007
II B.Tech. – I Sem
CSE (AI&ML)
ARTIFICIAL INTELLIGENCE

Question Bank

Course Code	23ES1305	**Year**	II	**Semester**	I
Course Category	Engineering Science	**Branch**	CSE (AI&ML)	**Course Type**	Theory
Credits	3	**L-T-P**	3-0-0	**Prerequisites**	Discrete Mathematical Structures, Probability and Statistics
Continuous Internal Evaluation	30	**Semester End Examination**	70	**Total Marks**	100

UNIT-I
Short Answer Questions(2 Marks Each)

QNO	QUESTION	CO	Blooms Level	Marks
1	Define Intelligence	**CO1**	**L2**	**2**
2	Define Artificial Intelligence	**CO1**	**L2**	**2**
3	Name one application of AI and Explain its significance	**CO1**	**L2**	**2**
4	Which assessment is used to test the machine	**CO1**	**L2**	**2**
5	Differentiate Rationality and Omni Science	**CO1**	**L2**	**2**
6	Define Agent function	**CO1**	**L2**	**2**
7	What are the operations in Vacuum cleaner world	**CO1**	**L2**	**2**
8	Why chess game is semi dynamic	**CO1**	**L2**	**2**
9	Expand and Define PEAS	**CO1**	**L2**	**2**
10	How to measure the performance of medical diagnosis system	**CO1**	**L2**	**2**
11	Define agent program.	**CO1**	**L2**	**2**
12	Define Problem	**CO1**	**L2**	**2**
13	Define abstraction	**CO1**	**L2**	**2**
14	**Describe the concept of rationality in AI?**	**CO1**	**L2**	**2**
15	**Define learning agent?**	**CO1**	**L2**	**2**
16	**Write two components of a problem-solving agent**	**CO1**	**L2**	**2**
17	**List the differences between strong AI and weak AI?**	**CO1**	**L2**	**2**

Long Answer Questions(10 Marks Each)

(Can have Sub Questions also)

QNO	QUESTION	CO	Blooms Level	Marks
1	Describe about various categories of AI.	**CO1**	**L2**	**10**
2	Outline the foundations of artificial intelligence.	**CO1**	**L2**	**10**
3	Discuss the Applications of AI	**CO1**	**L2**	**10**
4	Summarize the history of AI.	**CO1**	**L2**	**10**
5	Describe agent and how simple agent function for vacuum cleaner world works	**CO1**	**L2**	**10**
6	Describe rational agent and list out the differences of rationality and omniscience.	**CO1**	**L2**	**10**
7	Explain properties of task environment with example by using the concept of PEAS.	**CO1**	**L2**	**10**
8	Describe the role of PEAS in the task environment for an automated taxi.	**CO1**	**L2**	**10**
9	Outline the examples of agent types and their PEAS descriptions.	**CO1**	**L2**	**10**
10	Describe the following terms a. Fully Observable vs. Partial Observable b. Single agent vs. Multi agent c. Deterministic vs. stochastic. d. Episodic vs. sequential: e. Static vs. dynamic f. Discrete vs. continuous g. Known vs. unknown	**CO1**	**L2**	**10**
11	Describe the structure of Agent.	**CO1**	**L2**	**10**
12	Explain well - defined problems and solutions? How abstraction is useful while formulating problems?	**CO1**	**L2**	**10**
13 a)	**Explain the foundations of AI with suitable examples.**	**CO1**	**L2**	**5**
b)	**Describe the concept of rationality in intelligent agents. Why is it important.**	**CO1**	**L2**	**5**
14 a)	**Discuss the structure of an intelligent agent with a neat diagram.**	**CO1**	**L2**	**5**
b)	**Explain the role of Agent with vacuum world example.**	**CO1**	**L2**	**5**

UNIT-II

Short Answer Questions(2 Marks Each)

QNO	QUESTION	CO	Blooms Level	Marks
1	Define Parent Node with example.	**CO1**	**L2**	**2**
2	Define Search Strategy	**CO1**	**L2**	**2**
3	Describe cost in measuring problem solving performance	**CO1**	**L2**	**2**
4	Explain Zero-sum game with an example	**CO1**	**L2**	**2**
5	Define Game tree	**CO1**	**L2**	**2**
6	Explain Utility function to determine the Chess game	**CO1**	**L2**	**2**
7	Which state is a Terminal State	**CO1**	**L2**	**2**
8	Define Frontier	**CO1**	**L2**	**2**
9	Explain BFS and DFS in Uninformed search strategies	**CO1**	**L2**	**2**
10	Describe Search algorithms	**CO1**	**L2**	**2**
11	Define heuristic function	**CO1**	**L2**	**2**
12	Define local maxima	**CO1**	**L2**	**2**
13	**What is a search problem in AI?**	**CO1**	**L2**	**2**
14	**Define uniformed search strategies.**	**CO1**	**L2**	**2**
15	**Give one example of a real-life problem that can be modelled as a search problem?**	**CO1**	**L2**	**2**
16	**Write one difference between BFS and DFS**	**CO1**	**L2**	**2**

Long Answer Questions(10 Marks Each)

(Can have Sub Questions also)

QNO	QUESTION	CO	Blooms Level	MARKS
1	Implement Tree Search and Graph Search Algorithms to searching for solutions	**CO2**	**L3**	**10**
2	Illustrate Infrastructure for search algorithms and how to measure the performance of problem-solving algorithms.	**CO2**	**L3**	**10**
3	Use Breadth - first - search algorithm to evaluate Completeness, space complexity, time complexity, path cost.	**CO2**	**L3**	**10**
4	Illustrate uninformed cost search algorithm with suitable examples	**CO2**	**L3**	**10**
5	Apply Depth - first - search algorithm to evaluate Completeness, space complexity, time complexity, path cost.	**CO2**	**L3**	**10**
6	Apply Hill Climbing Algorithm to solve 8-Queeens Problem	**CO2**	**L3**	**10**
7	Apply A* search algorithm to minimizing the total estimated solution cost.	**CO2**	**L3**	**10**
8	Illustrate Mini Max Algorithm with an example	**CO2**	**L3**	**10**
9	Illustrate alpha - beta pruning with suitable example	**CO2**	**L3**	**10**

10	Analyze Tic Tac Toe problem with example	**CO4**	**L4**	**10**
11	Analyze BFS and DFS to evaluate Completeness, Space complexity ,time complexity and path cost	**CO4**	**L4**	**10**
12 a)	**Define state space search. Explain its role in solving AI problems**	**CO1**	**L2**	**5**
b)	**Describe the mini-max algorithm with an example of a two-player game**	**CO1**	**L2**	**5**
13 a)	**Discuss uninformed vs. informed search strategies with examples.**	**CO1**	**L2**	**5**
b)	**Explain alpha-beta pruning with a suitable example. How does it improve efficiency.**	**CO1**	**L2**	**5**

UNIT-III
Short Answer Questions(2 Marks Each)

QNO	QUESTION	CO	Blooms Level	Marks
1	Define Knowledge base	**CO1**	**L2**	**2**
2	Describe Knowledge representation language	**CO1**	**L2**	**2**
3	Give the differences between forward and backward chaining	**CO1**	**L2**	**2**
4	Define Inference rule	**CO1**	**L2**	**2**
5	Apply Inference rules for quantifiers	**CO1**	**L2**	**2**
6	Define Universal Instantiation	**CO1**	**L2**	**2**
7	How Skolemization is applied in knowledge representation	**CO1**	**L2**	**2**
8	Define semantics	**CO1**	**L2**	**2**
9	Give the differences between Propositional and First-Order Inference	**CO1**	**L2**	**2**
10	Define resolution	**CO1**	**L2**	**2**
11	Describe the role of lifting in knowledge representation	**CO1**	**L2**	**2**
12	Define Equality,tuple	**CO1**	**L2**	**2**
13	Define Ground sentences	**CO1**	**L2**	**2**
14	Define lifting lemma	**CO1**	**L2**	**2**
15	Define entailment	**CO1**	**L2**	**2**
16	Define Model	**CO1**	**L2**	**2**
17	Define Completeness	**CO1**	**L2**	**2**
18	Define Grounding	**CO1**	**L2**	**2**
19	Define Sound or truth preserving	**CO1**	**L2**	**2**
20	Operations in Knowledge base	**CO1**	**L2**	**2**
21	**Define propositional logic.**	**CO1**	**L2**	**2**
22	**Differentiate FOL with propositional logic?**	**CO1**	**L2**	**2**
23	**State one limitation of Propositional logic**	**CO1**	**L2**	**2**
24	**Give an example of an inference rule.**	**CO1**	**L2**	**2**

Long Answer Questions(10 Marks Each)
(Can have Sub Questions also)

QNO	QUESTION	CO	Blooms Level	MARKS
1	Explain logic and propositional logic in detail with an example.	**CO 1**	**L2**	**10**
2	Explain simple inference procedure with standard logical equivalences.	**CO 1**	**L2**	**10**
3	Illustrate the syntax and semantics of First Order Logic.	**CO 2**	**L3**	**10**
4	Illustrate the concept of First Order Logic	**CO 2**	**L3**	**10**
5	Consider a vocabulary with the following symbols: Occupation (p, o): Predicate. Person p has occupation o. Customer (p1, p2): Predicate. Person p1 is a customer of person p2. Boss (p1, p2): Predicate. Person p1 is a boss of person p2. Doctor, Surgeon, Lawyer , Actor : Constants denoting occupations. Emily, Joe: Constants denoting people. Use these symbols to write the following assertions in first-order logic: a. Emily is either a surgeon or a lawyer. b. Joe is an actor, but he also holds another job. c. All surgeons are doctors. d. Joe does not have a lawyer (i.e., is not a customer of any lawyer). e. Emily has a boss who is a lawyer. f. There exists a lawyer all of whose customers are doctors. g. Every surgeon has a lawyer.	**CO 2**	**L3**	**10**
6	Assuming predicates Parent(p, q) and Female(p) and constants Joan and Kevin, with the obvious meanings, express each of the following sentences in first-order logic. (You may use the abbreviation to mean “there exists exactly one.”) a. Joan has a daughter (possibly more than one, and possibly sons as well). b. Joan has exactly one daughter (but may have sons as well). c. Joan has exactly one child, a daughter. d. Joan and Kevin have exactly one child together. e. Joan has at least one child with Kevin, and no children with anyone else.	**CO 2**	**L3**	**10**
7	Illustrate Unification and Lifting in First Order Logic.	**CO 2**	**L3**	**10**
8	Analyze forward chaining algorithm with example	**CO 4**	**L4**	**10**
9	Analyze Backward chaining algorithm with example	**CO 4**	**L4**	**10**
10	Analyze Resolution algorithm with example.	**CO 4**	**L4**	**10**

11	Illustrate the structure of a Completeness proof for Resolution	**CO 2**	**L3**	**10**
12	**Explain knowledge based agents and Discuss the components involved in showing intelligence?**	**CO 1**	**L2**	**10**
13 a)	**Explain the connection between ∀ and ∃**	**CO 1**	**L2**	**5**
b)	**Differentiate between forward chaining and backward chaining.**	**CO 1**	**L2**	**5**
14 a)	Describe the architecture of a knowledge-based Agent with a neat diagram.	**CO 1**	**L2**	**5**
b)	Explain the working of a simple inference procedure with an example	**CO 1**	**L2**	**5**

UNIT-IV
Short Answer Questions(2 Marks Each)

QNO	QUESTION	CO	Blooms Level	Marks
1	Elaborate PDDl	**CO1**	**L2**	**2**
2	Explain classical planning with spare tire problem	**CO1**	**L2**	**2**
3	Describe planning graphs for heuristic estimation	**CO1**	**L2**	**2**
4	How to represent set of ground actions	**CO1**	**L2**	**2**
5	Define schema	**CO1**	**L2**	**2**
6	Define Levels in planning graphs	**CO1**	**L2**	**2**
7	Give the differences between Forward and Backward state space search	**CO1**	**L2**	**2**
8	Define optimistic description	**CO1**	**L2**	**2**
9	Describe planning approaches	**CO1**	**L2**	**2**
10	Define planning	**CO1**	**L2**	**2**
11	Define precondition, Effect	**CO1**	**L2**	**2**
12	Define Coordination	**CO1**	**L2**	**2**
13	Define Multi agent planning	**CO1**	**L2**	**2**
14	Define Hierarchical planning	**CO1**	**L2**	**2**
15	Define high level actions	**CO1**	**L2**	**2**
16	Define progression, regression	**CO1**	**L2**	**2**
17	Define Decomposition	**CO1**	**L2**	**2**
18	**What is state-space search?**	**CO1**	**L2**	**2**
19	**Define a planning graph**	**CO1**	**L2**	**2**
20	**What is the main goal of regression planning?**	**CO1**	**L2**	**2**

Long Answer Questions(10 Marks Each)
(Can have Sub Questions also)

QNO	QUESTION	CO	Blooms Level	MARKS
1	Apply Classical Planning with Air cargo Transport	CO 3	L3	10
2	Illustrate Classical Planning with The block world Examples.	CO 2	L3	10
3	Illustrate Forward (progression) state-space search Algorithm in detail	CO 2	L3	10
4	Illustrate Backward (regression) relevant-states search Algorithm in detail.	CO 2	L3	10
5	Illustrate Heuristics for planning.	CO 2	L3	10
6	Solve "The have cake and eat cake too" problem by using planning graphs concept.	CO 3	L3	10
7	Explain Graph plan algorithm.	CO 1	L2	10
8	Explain Hierarchical planning in detail.	CO 1	L2	10
9	Analyse doubles tennis problem by using Multi Agent Planning concept.	CO 4	L4	10
10	Analyze Cooperation and coordination Planning with multiple agents.	CO 4	L4	10
11	Analyze PDDL description for Spare Tire problem	CO 4	L4	10
12 a)	**Explain key characteristics that differentiate classical planning from other types of planning?**	CO 1	L2	5
b)	**Describe hierarchical planning advantages and provide an example of how it is used in a real-world application.**	CO 1	L2	5
13 a)	**Explain planning as state-space search with suitable examples.**	CO 1	L2	5
b)	**Discuss the importance of relevant states in regression planning.**	CO 1	L2	5
14 a)	**Explain the working of backward (regression) state-space search with a neat example.**	CO 1	L2	5
b)	**Write a detailed note on multi-agent planning. Explain the challenges and solutions.**	CO 1	L2	5

UNIT-V
Short Answer Questions(2 Marks Each

QNO	QUESTION	CO	Blooms Level	Marks
1	Define classification	CO1	L2	2
2	Define regression	CO1	L2	2

3	Define overfitting	**CO1**	**L2**	**2**
4	Explain Decision Trees with an example	**CO1**	**L2**	**2**
5	Describe Theory of Learning to restaurant problem	**CO1**	**L2**	**2**
6	Explain statistical learning to candy problem	**CO1**	**L2**	**2**
7	Describe Explanation based learning to extract general rules	**CO1**	**L2**	**2**
8	Define data matrix	**CO1**	**L2**	**2**
9	Define density estimation	**CO1**	**L2**	**2**
10	Explain Q-learning	**CO1**	**L2**	**2**
11	Define Utility function	**CO1**	**L2**	**2**
12	Define Learning curve	**CO1**	**L2**	**2**
13	**Define explanation-based learning?**	**CO1**	**L2**	**2**
14	**Define the term "reward" in reinforcement learning.**	**CO1**	**L2**	**2**
15	**What is overfitting in decision trees.**	**CO1**	**L2**	**2**
16	**Differentiate between positive and negative reinforcement**	**CO1**	**L2**	**2**

Long Answer Questions(10 Marks Each)

(Can have Sub Questions also)

QNO	QUESTION	CO	Blooms Level	MARKS
1	Explain different Forms of Learning.	**CO 1**	**L2**	**10**
2	Analyze Inducing Decision Trees with a case study	**CO 4**	**L4**	**10**
3	Analyze Decision Tree Learning algorithm on restaurant.	**CO 4**	**L4**	**10**
4	Analyze how to evaluate the PAC Learning with suitable case study	**CO 4**	**L4**	**10**
5	Explain the theory of learning with examples	**CO 1**	**L2**	**10**
6	Explain the following terms: a. Univariate linear regression b. Multivariate linear regression	**CO 1**	**L2**	**10**
7	Explain Learning an action utility function	**CO 1**	**L2**	**10**
8	Explain Statistical Learning methods	**CO 1**	**L2**	**10**
9	Apply Passive Reinforcement Learning with an example	**CO 3**	**L3**	**10**
10	Explain Active Reinforcement Learning with an example	**CO 1**	**L2**	**10**
11	Explain Explanation based learning with example	**CO 1**	**L2**	**10**
12 a)	**Analyze the main principles that guide the learning process in AI systems?**	**CO 4**	**L4**	**5**

b)	**Explain the decision tree learning with an example.**	**CO 1**	**L2**	**5**
13	**Explain the theory of learning in AI. Discuss the PAC (Probably Approximately Correct) learning framework.**	**CO 1**	**L2**	**10**
14	**Write a detailed note on statistical learning methods with suitable examples.**	**CO 1**	**L2**	**10**

Course Coordinator:

Module Coordinator: (Signature of HOD)